

Event Management & Registration System

1. Project Overview

Build a simple role-based Event Management & Registration System where organizers can create and manage events and attendees can browse and register for events. The project is designed as a beginner-to-intermediate intern assessment and focuses on authentication, role-based authorization, CRUD operations, many-to-many relationships, validations, basic dashboards, and simple business rules.

2. Roles

- Admin
- Event Organizer
- Attendee

3. Core Modules

Authentication & Role Management

- User signup and login.
- Logout and password reset.
- Role-based authorization.
- Admin can activate/deactivate users.
- Users can update their own profile.

Event Management

- Event Organizers can create events.
- Organizers can edit and update their own events.
- Organizers can view event details.
- Organizers can cancel events.
- Organizers can see the attendees registered for their events.
- Admin can view, edit, update, and delete any event.
- Attendees can view published events.

Event Attributes

- Title
- Description
- Event date
- Start time
- End time
- Location
- Capacity

- Category
- Status
- Organizer
- Optional event image

Event Categories

- Admin can create, edit, and delete categories.
- The organizer selects a category while creating an event.
- Examples: Workshop, Seminar, Conference, Sports, Meetup, Training.

Event Registration

- Attendees can register for a published event.
- Attendees can view their registered events.
- Attendees can cancel their registration.
- Organizers can view registered attendees for their events.
- Admin can view registrations across the system.

Event Capacity & Validation

- Events must have a positive capacity.
- Registration should be blocked when the event is full.
- An attendee cannot register for the same event more than once.
- An attendee cannot register for a cancelled or completed event.
- Registration should be allowed only for published events.
- After an attendee cancels, the available capacity should increase.

Event Status Management

- Draft → Published → Completed → Cancelled.
- Organizers can publish a draft event.
- Organizers can cancel a published event.
- Completed events should no longer accept registrations.
- Cancelled events should no longer accept registrations.
- Admin can manage event status when necessary.

Search & Filtering

- Attendees can search events by title or description.
- Filter events by category.
- Filter by date.
- Filter by event status.
- Optional sorting by date, title, or available seats.

Role-Based Dashboards

- Each role should have a simple dashboard showing information relevant to them.
- Dashboard statistics should be calculated from actual application data.

4. Dashboard Requirements

Admin Dashboard

- Total users.
- Total organizers.
- Total attendees.
- Total events.
- Published events.
- Completed events.
- Cancelled events.
- Total registrations.
- Recent events and registrations.

Event Organizer Dashboard

- Total events created by the organizer.
- Draft events.
- Published events.
- Completed/cancelled events.
- Total registrations across their events.
- Upcoming events.
- Recent registrations.

Attendee Dashboard

- Total registered events.
- Upcoming registered events.
- Completed events attended/registered for.
- Recently registered events.
- Available upcoming events.

5. Many-to-Many Relationship

The primary database concept being evaluated is the relationship between Users/Attendees and Events. An attendee can register for many events, and an event can have many attendees.

User/Attendee ↔ Event

This relationship should be represented through a Registration/Enrollment model.

Suggested Registration attributes:

- Attendee/User
- Event
- Registration status
- Registration date
- Optional cancellation date

6. Registration Status

- Registered
- Cancelled

A cancelled registration should not count toward the event's current attendee capacity.

7. Important Business Rules

- Only attendees can register for events.
- Only published events can receive registrations.
- An attendee cannot register for the same event twice.
- An event cannot exceed its defined capacity.
- Cancelled registrations should free up a seat.
- Completed and cancelled events cannot receive new registrations.
- An organizer can manage only their own events unless they have Admin privileges.
- An attendee can only cancel their own registration.
- Event capacity cannot be zero or negative.
- Event start date/time should not be after its end date/time.
- Event date should not be invalid or incomplete.
- Deleting an event with registrations should be handled carefully; soft deletion or cancellation can be used instead.

8. Event Image Upload (Optional)

- Organizer can upload an event cover image.
- Display the image on the event details page.
- Use Active Storage or an equivalent cloud storage solution.
- Image upload should be optional so it does not block interns who are still learning file handling.

9. Optional / Bonus AI Module

AI Event Assistant

- Allow users to ask simple questions about available events.
- Example: 'What workshops are available this month?'
- Example: 'Show me events in Lahore.'
- Example: 'Which events still have seats available?'
- Example: 'What events am I registered for?'
- The AI should only access information the authenticated user is authorized to see.
- The AI should not be responsible for directly changing registrations.

10. Suggested Data Model

User: name, email, password, role, status, profile image (optional), timestamps

Event: title, description, date, start time, end time, location, capacity, category, organizer, status, image (optional), timestamps

Category: name, description, status, timestamps

Registration: attendee/user, event, status, registered at, cancelled at (optional), timestamps

Notification (optional): user, title, message, type, read status, timestamp

11. Key Relationships

- User has many Events as an Organizer.
- Event belongs to an Organizer/User.
- Event belongs to a Category.
- Event has many Registrations.
- User/Attendee has many Registrations.
- User/Attendee has many Events through Registrations.
- Event has many Attendees through Registrations.
- User has many Notifications if notifications are implemented.

12. Important Scenarios to Test

- Event capacity is 50 and 50 attendees are registered: the next registration must fail.
- Attendee cancels registration: one seat becomes available.
- Attendee attempts to register twice: second registration must fail.
- Attendee attempts to register for a draft event: registration must fail.
- Attendee attempts to register for a cancelled event: registration must fail.
- Attendee attempts to register for a completed event: registration must fail.
- Organizer attempts to edit another organizer's event: unauthorized.
- Attendee attempts to cancel another attendee's registration: unauthorized.
- Event organizer creates an event with capacity 0 or a negative capacity: validation error.
- Event start time is later than end time: validation error.
- Admin can access and manage all events.
- Cancelled registrations should not be included in the current attendee count.

13. Evaluation Focus

- Correct authentication and role-based authorization.
- Clean database relationships, especially many-to-many registration.
- Correct CRUD implementation.
- Backend validation and business rules.
- Capacity calculation and duplicate registration prevention.
- Correct handling of event status.
- Authorization around organizer-owned events.
- Search, filtering, and pagination.
- Simple, usable dashboard design.
- Clean API and frontend structure.
- Meaningful error handling.
- Testing of common edge cases.
- Maintainable and readable code.